

Network Design Document

Meridian State University

A Scalable, Secure, Hybrid Campus Network Architecture

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Course: Enterprise System Architecture

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Date: June 2026

1. Executive Summary

Meridian State University is a fictional public university with 20,000 students and 3,000 faculty and staff across twelve buildings on one contiguous campus, and this document is the network design for that environment. The brief was to build something that handles the full mix of services a modern campus runs on, wired desks, dense Wi-Fi, VoIP phones, IoT and building automation, physical security cameras, a server farm, isolated guest access, and eduroam for visiting researchers, while staying scalable as enrollment grows, available when individual components fail, and tight enough that a compromised student laptop or hacked camera cannot reach the administrative systems that actually run the university.

The architecture is a three-tier hierarchy of core, distribution, and access switching, sitting behind a redundant internet edge and tied to a data-center block in the Keating building. The campus is segmented into fourteen purpose-built VLANs and addressed out of the private 10.0.0.0/8 space using a hierarchical schema, where the second octet identifies the building and the third identifies the VLAN within it, which keeps routing tables small at the core and gives every site obvious room to grow. Redundancy is engineered into every layer rather than added on top of one of them, with redundant core switches, redundant distribution pairs per building, dual uplinks from every access switch, HSRP at the gateway, dual ISPs at the edge, and UPS plus generator behind the power feed. A controller-managed Wi-Fi 6/6E system covers the buildings, and a defense-in-depth security stack of 802.1X at the port, default-deny between VLANs, next-generation firewalls at the edge, encryption in transit and at rest, and centralized logging enforces a zero-trust posture from the user's port inward. The campus then connects to AWS over a dedicated Direct Connect link so that identity, hosted services, logging, and disaster recovery live in the cloud while the wires, switches, and access points stay on campus. That hybrid posture is the same model NIST describes in Special Publication 800-145 and the baseline AWS treats as the default in its own enterprise architecture guidance, applied here at the scale of a mid-sized public university.

2. Project Requirements and Design Goals

The user population is large and uneven. Residential buildings are the densest environments on campus because every student in a dorm room is online at the same time on at least two or three devices, while academic and research buildings carry heavy wireless and lab loads that spike around scheduled class blocks. The building inventory below set the scope and capacity planning for the rest of the design.

Building	Primary Use	Approx. Peak Devices
Halsey, Brooks, Tanaka Halls	Classrooms and teaching labs	800–1,200 each
Wexler Science Center	Research and instructional labs	~1,500
Cordova Engineering	Labs and maker space	~1,200
Aldridge Library	Study space (wireless-dense)	~2,000
Sutter, Vance, Paulson Halls	Residence halls	~6,000 each

Meridian Union	Dining, recreation, events	~2,500
Bell Administration	Staff and administrative offices	~600
Keating Data Center	Core infrastructure and servers	Not user-facing

Table 1. Building inventory and approximate peak device counts.

Capacity numbers are only half of the picture. The design also had to satisfy a set of non-functional goals that ended up driving most of the architectural decisions downstream.

- Scalability. New buildings, new users, and new device classes need to plug into the existing structure rather than force a redesign every few years.
- High availability. Every layer of the network has a redundant path, and a failure in one component does not interrupt service in the rest.
- Security and compliance. Segmentation, port-level authentication, and centralized logging strong enough to protect administrative, financial, and research systems.
- Performance. Fiber backbone, right-sized broadcast domains, and route summarization to keep forwarding decisions fast at peak load.
- Manageability. Switching, wireless, and identity all managed from central, cloud-assisted control planes rather than per-device.

Beyond user counts, the network has to carry the full mix of services a modern campus depends on: wired and Wi-Fi access, VoIP telephony, building automation and IoT, physical security cameras, the server and data-center block, an isolated guest network, eduroam for visiting academics, and outbound traffic through redundant internet providers.

3. Network Architecture Overview

At 23,000 users and somewhere north of 40,000 concurrent devices spread across twelve buildings, MSU is well past the size where a flat or collapsed-core network still works in practice. The design therefore uses the three-tier hierarchical model that has been the standard for enterprise campus networks since Cisco first formalized it, with two specialized blocks added on top: a data-center block in Keating that holds the servers, controllers, and identity services, and an internet-edge block that terminates the ISP connections and the AWS Direct Connect link.

3.1 The Three Tiers

Each tier has a job. The access layer sits in the wiring closets on every floor of every building and is where end devices actually plug in or associate, so it provides the physical ports, Power over Ethernet for phones, cameras, and access points, and the first authentication check through 802.1X and port security. The distribution layer aggregates the access switches inside one building, performs inter-VLAN routing for that building, and is where most of the security and access-control policy lives, deployed as a redundant pair in every building so the loss of one switch does not isolate the building. The core layer sits in the Keating Data Center as a redundant pair of high-speed switches whose only job is to forward

traffic between distribution blocks and the edge as fast and predictably as possible, deliberately stripped of policy so that the core stays a forwarding plane rather than a place where bugs and exceptions accumulate.

Separating concerns this way solves three problems at once: a failure inside one building's distribution block stays inside that building because the core does not depend on it, a thirteenth or fourteenth building is added by standing up a new distribution block and attaching it to the existing core, and troubleshooting becomes layer-by-layer rather than a hunt through a tangle because each layer has a defined responsibility. That logic is what AWS calls out in its enterprise architecture guidance when it describes low-latency, high-throughput, redundantly connected infrastructure as the foundation reliable services are built on, and the three-tier model is the campus expression of the same idea.

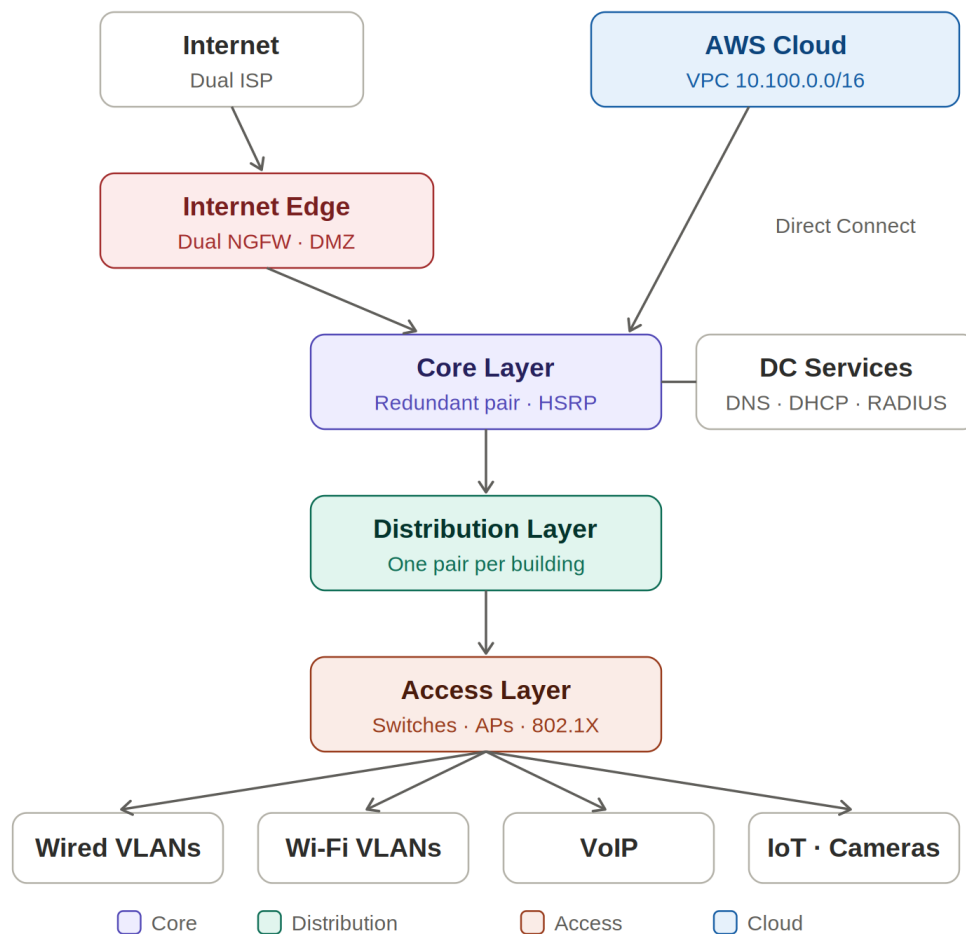


Figure 1. Logical network architecture: three-tier hierarchy with internet edge and AWS cloud integration.

4. VLAN Segmentation Plan

One physical network is sliced into many logically isolated broadcast domains using Virtual LANs. The point of doing it is more about containment than convenience: by default no VLAN may talk to another unless an explicit rule allows it, which means a malware infection on a student device cannot reach the administrative servers, a hacked camera cannot scan the finance subnet, and a compromised IoT controller stays in its building automation zone. VLANs are grouped by role and sensitivity and numbered in functional blocks (1xx for staff, 2xx for students, 3xx for wireless, 4xx for voice, 5xx for OT) so that the plan stays readable as it grows.

VLAN	Name	Members	Security Posture
10	Management	Switch, AP, and controller administration	Most restricted; admins only
20	Data Center / Servers	Application and file servers	Tightly restricted
30	Network Services	DNS, DHCP, RADIUS	Restricted
110	Faculty/Staff Wired	Office desktops and wired staff	Standard internal
120	Admin / Finance	HR, payroll, bursar systems	High; treated as a PCI-style zone
210	Student Wired	Computer labs, library wired	Isolated from staff
220	Residence Halls	Dorm room wired connections	Isolated; campus + internet
310	Faculty/Staff Wi-Fi	Staff wireless devices	Standard internal
320	Student Wi-Fi	Student wireless devices	Isolated from staff
330	Guest Wi-Fi	Visitors	Internet-only; fully walled off
340	eduroam	Roaming academic Wi-Fi	Authenticated; isolated
410	Voice (VoIP)	Desk phones	Separated for QoS and security
510	Building Automation	HVAC, lighting, badge readers	Locked to controllers
520	Physical Security	Cameras, door access	Locked down; no internet

Table 2. Campus VLAN plan.

5. IP Addressing Schema

Addressing comes out of the private 10.0.0.0/8 block, which gives roughly 16 million usable addresses and a lot of room to grow into. The schema is hierarchical and reads like a postal address: the second octet identifies the building, the third identifies the subnet or VLAN inside that building, and the fourth identifies the host. That hierarchy is what lets the core hold one summarized route per building instead of hundreds of individual subnet routes, which is the main reason hierarchical addressing scales gracefully at this size.

5.1 Campus Address Blocks

Each building gets its own /16, which is far more than required by design. 65,000-plus addresses per building is generous enough that no future split, IoT expansion, or extra wireless density runs out of space, and the surplus makes a clean /16 allocation per building easier to justify than trying to right-size a smaller block now and migrate later.

#	Building / Zone	Address Block
1–3	Halsey, Brooks, Tanaka Halls	10.1.0.0/16 – 10.3.0.0/16
4	Wexler Science Center	10.4.0.0/16
5	Cordova Engineering	10.5.0.0/16
6	Aldridge Library	10.6.0.0/16
7–9	Residence Halls	10.7.0.0/16 – 10.9.0.0/16
10	Meridian Union	10.10.0.0/16
11	Bell Administration	10.11.0.0/16
12	Keating Data Center	10.12.0.0/16
—	AWS Cloud (VPC)	10.100.0.0/16
—	Reserved for growth	10.200.0.0/16

Table 3. Campus address block allocation.

5.2 Subnetting Within a Building

Inside each building's /16, every VLAN gets a right-sized subnet using Variable Length Subnet Masking. Subnet sizes are matched to actual demand rather than handed out uniformly, so management gets a /24 and student Wi-Fi gets a /22 because that is what those populations look like in practice. Tanaka Hall (10.3.0.0/16) is the worked example.

VLAN	Name	Subnet	Usable	Gateway
10	Management	10.3.0.0/24	254	10.3.0.1
110	Faculty/Staff Wired	10.3.1.0/24	254	10.3.1.1

210	Student Wired	10.3.2.0/24	254	10.3.2.1
410	Voice (VoIP)	10.3.4.0/24	254	10.3.4.1
510	Building Automation	10.3.5.0/24	254	10.3.5.1
520	Physical Security	10.3.6.0/24	254	10.3.6.1
310	Faculty/Staff Wi-Fi	10.3.8.0/23	510	10.3.8.1
320	Student Wi-Fi	10.3.16.0/22	1,022	10.3.16.1

Table 4. Tanaka Hall (Building 3) VLSM subnetting example.

5.3 Residence Hall Subnetting

Residence halls are the special case. Each holds roughly 6,000 devices once laptops, phones, gaming consoles, smart TVs, and the surprising number of IoT gadgets students bring with them are added up, and putting all of those in one subnet would create a broadcast domain large enough to chew through switch CPU and degrade Wi-Fi roaming. The dorm network is subdivided by floor instead, with each floor on its own /23 of 510 usable addresses (10.7.16.0/23 for floor one, 10.7.18.0/23 for floor two, and so on), which keeps the broadcast traffic per floor manageable and makes localized problems easier to diagnose.

5.4 Cloud Address Coordination

The AWS Virtual Private Cloud is deliberately given a non-overlapping block, 10.100.0.0/16, that does not collide with any of the on-campus building ranges. Non-overlapping address space is the prerequisite for hybrid networking, because if the campus side and the VPC side ever overlapped, every packet crossing the Direct Connect link would need network address translation and the routing would stop being deterministic. With 10.1 through 10.12 on campus and 10.100 in the cloud, the two halves route to each other as one network without any of that.

6. Physical Design

The logical hierarchy maps onto physical equipment distributed across the campus in a fairly direct way. Every floor of every building has an Intermediate Distribution Frame, which is the wiring closet that holds the access switches users actually connect to. Every building has a Main Distribution Frame, usually on the ground floor, that holds the redundant distribution-switch pair. The Keating Data Center holds the core switches, the wireless controllers, the internet-edge equipment, the cloud-connectivity router, and the server racks that everything is ultimately talking to.

Access switches deliver Power over Ethernet to anything that lives at the edge of the network, which means phones, cameras, and access points pull both data and electrical power from one cable rather than needing a separate electrical supply at every drop. The physical media are chosen by run length and the performance the link has to carry, as summarized below.

Link	Medium	Rationale
Desk to access switch	Copper (Cat6)	Inexpensive and sufficient for short runs
Floor IDF to building MDF	Multimode fiber	High speed, immune to electrical interference
Building MDF to core	Single-mode fiber	Carries the backbone across campus distances

Table 5. Physical media selected per link type.

Figure 2 shows the physical topology end to end: Keating as the hub, fed by both internet providers and the Direct Connect link to AWS, with a single-mode fiber backbone running out to each building where the MDF hands off to the per-floor IDFs.

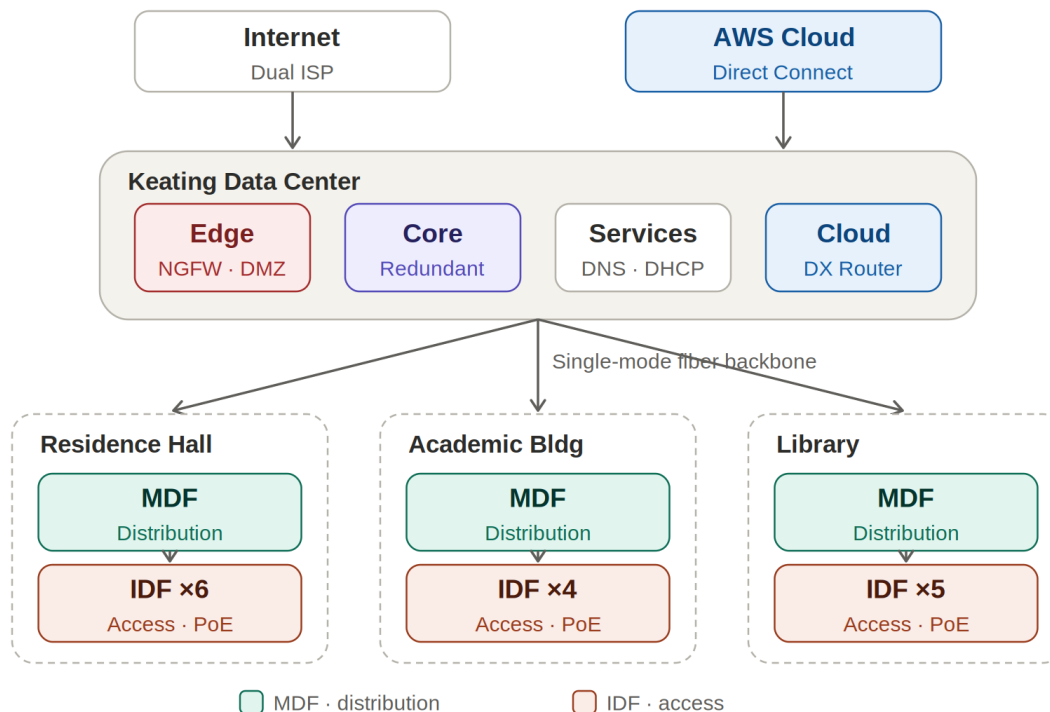


Figure 2. Physical network topology: data-center hub, fiber backbone, and per-building MDF/IDF structure.

7. Redundancy and High Availability

High availability is not a single feature in this design, it is a posture applied at every layer. Whenever a component fails it has a standby that takes over fast enough that users do not notice, and "every layer" matters because a network that is redundant at the core but single-homed at the access switch only survives the failures that happen to occur in the right place. The mechanisms used are:

- Duplicated devices at every tier: redundant core switches, redundant distribution pairs in every building, and dual power supplies in critical equipment.
- Dual uplinks from every access switch into both distribution switches, so that a single cable cut or port failure does not isolate the closet.
- First-hop redundancy with HSRP. The distribution pair shares one virtual gateway address, and if the active switch goes down the standby takes over that address in about a second, which end devices never see.
- EtherChannel link bundling. Multiple physical links are bonded into one logical link, which both adds bandwidth and survives the loss of any single member cable.
- Spanning Tree Protocol to prevent loops in the redundant topology while keeping the backup paths warm.
- Dual internet providers at the edge, so a single ISP outage does not disconnect the campus.
- Layered power protection: dual feeds from an uninterruptible power supply backed by a standby generator, so the lights stay on through both short utility blips and longer outages.

8. Wireless Infrastructure

Most users associate over Wi-Fi rather than plug into a wall, so the wireless system is treated as a primary design element rather than an overlay. A redundant pair of Wireless LAN Controllers in the Keating Data Center centrally manages every access point on campus, which means a single policy change propagates everywhere in one push and the wireless service survives the loss of either controller. Access points are Wi-Fi 6/6E units powered by PoE off the access switches and placed by density rather than by floor plan, so coverage is light in offices and heavy in the library and the residence halls where simultaneous demand is the highest.

Every SSID is mapped to a VLAN from the segmentation plan, so the student SSID lands on the student Wi-Fi VLAN, faculty lands on the staff VLAN, guests get the walled-off guest VLAN, and eduroam goes to its own isolated VLAN for visiting academics. The authentication step is 802.1X against the RADIUS service: when a user associates, the controller hands the credentials to RADIUS, RADIUS checks them against the cloud-backed identity directory, and the user is dropped on the correct VLAN automatically. That identity check at the moment of joining the air is what keeps the wireless side consistent with the zero-trust posture the rest of the network uses.

9. Security Architecture and Policies

Security in this design follows defense in depth, which means stacking independent controls so that defeating any one of them does not give an attacker a free path to the next. Most of the controls are already built into the architectural choices described above, so this section is mostly about the policies that govern them.

9.1 Network Access Control

No device gets an address until it authenticates. Using 802.1X, the access port stays closed at the data-link layer until the RADIUS service, which is itself backed by cloud identity, confirms the user; once confirmed the port opens and drops the device onto the right VLAN. Unknown or failing devices land on either a deny rule or a quarantine VLAN with no access to anything sensitive. The point is that nothing is trusted because of where it is plugged in, identity has to be proven first, which is the same principle that drives cloud identity and access management.

9.2 Segmentation Policy

Inter-VLAN traffic is default-deny. A flow is permitted only if there is a documented business reason for it, and sensitive flows are logged in addition to being allowed. The table below is a representative subset of the full policy.

Source	Destination	Allowed	Rationale
Student Wi-Fi (320)	Internet	Yes	Normal usage
Student Wi-Fi (320)	Admin/Finance (120)	No	No business need; blocked
Guest (330)	Any internal VLAN	No	Internet-only by design
Cameras (520)	Camera server only	Yes	Locked to a single destination
Cameras (520)	Internet	No	IoT devices contained
Admin/Finance (120)	Data Center servers (20)	Yes (logged)	Required but monitored

Table 6. Excerpt of the inter-VLAN segmentation policy.

9.3 Perimeter, Encryption, and Monitoring

At the internet edge, a redundant pair of next-generation firewalls separates the campus from the outside world. They inspect outbound traffic and admit inbound traffic only to systems that have explicitly been published, so public-facing services like the university website and public DNS live in a demilitarized zone rather than inside the trust zone, which means even a fully compromised public web server cannot use that compromise to reach internal systems. This is the same DMZ pattern Microsoft documents in its cloud-networking guidance for enterprise architects.

Sensitive data is encrypted both in transit using TLS and at rest using a managed key-management service, so that key material is not stored next to the data it protects. Logging is centralized: switch, firewall, and authentication logs are aggregated and analyzed in one place, and cloud-side activity is captured by AWS CloudTrail for who-did-what auditing and CloudWatch for system health. An intrusion detection and prevention system watches traffic for known attack patterns and can either alert or automatically block. Taken together, these controls express the network's zero-trust philosophy in five

places at once: identity at the port, segmentation between zones, inspection at the perimeter, encryption of the data itself, and centralized monitoring across everything.

10. Hybrid Cloud Integration

A campus network is unavoidably physical because the users, wall ports, and access points have to be where the humans are. What does not have to live on campus anymore is the management plane, the identity service, the hosted applications, and the disaster-recovery target, and there is a real cost to keeping all of that on-premises now that the cloud equivalents are mature and cheaper to run. MSU is designed as a hybrid enterprise architecture for that reason: the on-premises network is paired with AWS for management, identity, hosted services, and disaster recovery. This is the hybrid deployment model NIST defines in SP 800-145 and the same baseline AWS treats as the default in its own enterprise architecture guidance.

The campus-to-cloud link is AWS Direct Connect, a dedicated, private, high-throughput connection rather than a tunnel over the public internet. As Microsoft's cloud-networking guidance notes for the equivalent ExpressRoute service on the Azure side, the value of a dedicated link is that the cloud-hosted resources behave as though they are on the organization's own network, with predictable performance and stronger isolation than best-effort internet routing. The external services and their AWS counterparts are summarized below.

Service	How It Is Handled	AWS Component
Identity / login	Cloud directory feeds RADIUS for 802.1X	IAM Identity Center
DNS	Internal DNS plus public campus domain	Amazon Route 53
Hosted applications	LMS, email, and storage delivered as SaaS	Hosted in / fronted by AWS
Backup and disaster recovery	Critical data replicated off-site	Amazon S3 (cross-region)
Campus-to-cloud link	Dedicated private connection	AWS Direct Connect

Table 7. External services and their AWS counterparts.

The disaster-recovery posture is the biggest single benefit of the hybrid model. Because the critical systems and their data are continuously replicated to AWS, a catastrophic loss of the Keating Data Center, whether from fire, flood, or extended power failure, would not end university operations: the services come back up out of the cloud and traffic eventually returns to campus once Keating is rebuilt. Security governance also benefits from cloud tooling, since services like AWS Trusted Advisor surface availability, security, and best-practice findings on a continuous basis, which is the kind of automated review capability that is hard to replicate in a purely on-premises environment.

11. Performance and Scalability

Performance comes from a few choices reinforcing each other. The single-mode fiber backbone with EtherChannel-bundled uplinks gives the inter-building and core links the bandwidth and headroom they need at peak. Broadcast domains are kept small by VLSM-sized subnets and by the per-floor split inside residence halls, which keeps broadcast overhead and latency low even when an entire dorm is online at once. Hierarchical addressing then enables route summarization at the core, which shrinks the forwarding tables the core has to walk and keeps lookups fast under load.

Scalability is a structural property of the design rather than something added on top of it. Each building's /16 allocation gives so much address headroom that no realistic expansion runs out of subnets, and the three-tier model lets new buildings come online as self-contained distribution blocks attached to the existing core without touching the buildings already in service. The hybrid cloud dimension extends scaling further still: elastic AWS capacity absorbs burst demand and hosts new services without buying on-premises hardware, which is the kind of agility AWS and cloud-native guidance both cite as the main reason to adopt cloud in the first place. Continuous performance monitoring, both on the campus side and through cloud tooling, keeps capacity tuned to actual demand over time rather than to whatever the design assumed on day one.

12. Conclusion

The design meets every goal that was set at the start. It scales because the addressing is hierarchical and the architecture is modular at the building level, so growth is a matter of attaching new distribution blocks rather than rebuilding the core. It is secure because the network is segmented by role at the VLAN layer, authenticated at the port, defended at the perimeter, encrypted in transit and at rest, and logged centrally for review. It is highly available because every layer carries redundancy as a default, from dual access uplinks at the closet all the way out to dual ISPs and a backup generator. And by integrating the on-premises campus with AWS over a dedicated Direct Connect link, the design becomes a real hybrid enterprise architecture rather than just a campus LAN, with cloud identity, cloud-hosted services, centralized logging, and disaster recovery all part of the same fabric. The result is a network sized for 23,000 users today and clearly able to grow with the institution into whatever it needs to be five and ten years from now.

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